

School of Computer Science and Engineering

Object Oriented Programming with Java

Assignment-1

1. Demonstrate the use of this and super keywords in resolving constructor chaining and variable shadowing. Analyze their role in multi-level inheritance
2. Define a class Candidate in JAVA with following description:
Data Members like Roll_No , Name, Score and remarks .
Member functions are: AssignREM() to assign remarks as per score obtained by candidate score range as given below,

Score	Remarks
≥ 50	Selected
< 50	Not selected

ENTER(): to allow the user to enter values for Roll_No, Name, Score and call function AssignREM() to assign remarks.

Display() : to allow the user to display all data members.

3. Develop a Java program to define class called Book with data members bookNo, bookTitle, bookPrice and create a member functions to read and display 5 book details. Create Book objects and implement the same.
4. Implement a Java Program for the scenario, **SchoolMember** is a base class which has properties like name, age and address methods like **showDetails()** to display these properties. There are 3 subclass
Teacher: with additional properties like subjectSpecialization and methods like **teach()**.
Student: with properties such as gradeLevel and methods like **study()**.
Staff might have properties like department and methods like **work()**.
Display the details of the Teacher, Student and Staff.

5. JAVA provides different types of variables within class.

```
class Var{  
    int a; static int b=20;  
    Var(int x) {  
        a=x; }  
}
```

Differentiate between different types of variables by considering the above example.

6. Write a Java program to create an interface Vehicle with a method start(). Implement this interface in two classes Car and Bike. Display the starting message for each vehicle.
7. Explain the visibility of class members when accessed within the same package and different packages.
8. Write a Java program to overload a method add() to calculate sum of two integers, three integers, and two double values.

9. Write a program to demonstrate runtime polymorphism using method overriding.
10. Create two packages and show how a class from one package can be accessed in another using import.
11. A school wants to maintain information about its staff. There are **two types of staff** – **Teaching** and **Non-Teaching**.
 - Every staff member has a name and ID.
 - Teaching staff has subject and number of classes.
 - Non-Teaching staff has department and work hours.

Write a Java program using inheritance to display details of both types of staff.

12. Write a Java program showing how a subclass can access methods of an abstract superclass using super.
13. Create an interface for a banking system with methods deposit() and withdraw(). Implement it in Customer class.
14. Illustrate how to create and use a user-defined package in Java with a suitable program.
15. Define an interface "Area" which should contain a method named area (); Implement this interface in rectangle & triangle classes to find their area. Demonstrate the above implementation by writing a main method in a new class called "DemoInterfaceArea".